

K472 — Three CIs Absorb Cal #344 onto Own Work; Selection Rules Are Prior + Blind + Cal-#344-Compliant; Bulk Anomalous Dimension on $H^2(D_{IV}^5)$ Is the Real Open Computation

Date: 2026-06-22 (Monday, ~12:15 EDT date-verified) · Auditor: Keeper · Inputs: Lyra F274 selection rules; Elie Toy 4308 walk-back + Toy 4310 derived selection-rule table; Grace self-applied Cal #344 to her K469 nuclei bridge with $N_{\max}=137$ trap explicitly flagged

Verdict — substantive Cal #344-discipline cascade across THREE CIs onto their own Monday-morning work; selection-rule structure is prior + blind + SOLID; the real open computation is the bulk anomalous dimension on $H^2(D_{IV}^5)$

Count holds at 4 of 26. The substantively distinct content this turn: the team applied Cal #344 to their own morning work, with self-corrections converging on the right framing.

Landing 1 — Elie walks back his own Toy 4308 — PASS at SOLID discipline event

The walk-back

Elie applied Cal #344 to his own 4308 complexity-ordering work:

“Cal #344 caught Keeper for post-hoc class-attribution (assigning operator classes after seeing which channel closed, then calling it a prediction). Two pieces of my 4308 had the same problem, so I applied the blade to my own work — the way Keeper, Grace, and Lyra each did this weekend”

Two specific retractions:

1. “ 0^{-+} = color-cubic $d^{\{abc\}}$ ” was wrong physics. The real 0^{-+} glueball is $\text{Tr}(F \cdot \tilde{F})$ — same color contraction as 0^{++} , differing only by spacetime Hodge dual. Lyra’s correction (her own F-note this morning) is decisive. The $d^{\{abc\}}$ operator IS valid Hermitian cubic, just the wrong object for 0^{-+} .
2. “Complexity orders the masses” correlation collapses. It rested on the misidentification PLUS a complexity metric Elie tuned by hand to make the order come out. That’s the exact post-hoc pattern Cal #344 caught Keeper for. Dropped.

What survives is better than what dropped

Lyra’s factorization (which Elie verified computationally) is the real unlock. The $d^{\{abc\}}$ cubic finding (anti-Hermitian $f^{\{abc\}}$ ruled out; symmetric $d^{\{abc\}}$ Hermitian) stands as operator-algebra fact. The qualitative ordering match (1^{+-} heaviest because derivative-required, canonical dim 6 vs 4) ALSO stands as a blind consistency — fixed before looking at lattice numbers.

Landing 2 — Grace self-applies Cal #344 to her K469 nuclei bridge — PASS at SOLID discipline event

The acknowledgment

Grace explicitly: *“Cal #344’s catch on Keeper applies squarely to me too — my K469 nuclei bridge was starting down the same cascade.”*

Substantive corrections to her K469 framing

1. End-of-periodic-table requires TWO unproven steps: (a) finite class set, (b) class-set bounds shells. “BST’s magic-number shell tower is infinite: it’s already in the program that the next magic number after 126 is 184, and the cumulative Wallach K-type structure keeps going past that.”

2. $N_{\max} = 137$ trap explicitly flagged. Grace pinned the exact same $N_{\max}$ -vs-Bohr-137 concern I flagged in K470: > “BST has $N_{\max} = 137 = N_c^3 \cdot n_C + \text{rank}$ as a substrate primary. And the famous Bohr/Feynman argument gives $Z \approx 137$ as where the inner electron goes light-speed. Two readings of 137. ... I can’t tell which without the substrate-mechanism, and saying ‘BST has 137, physics has 137, therefore the table ends at 137’ is precisely the post-hoc cascade Cal caught yesterday.”

What would earn the prediction: a substrate shell-counting argument that forces the cutoff from operator-class structure, NOT borrowed from the Bohr argument.

3. “Wrong glueball for shell” question = the sharpest falsifier Casey raised IF taxonomy survives blind test.

Landing 3 — Selection-rule structure is PRIOR + DERIVED + BLIND — PASS at SOLID-substantive (Cal #344 compliant)

What Lyra F274 + Elie 4310 + Grace independently confirmed

Two derived selection rules for 2-gluon glueballs:

SR1 — Gauge invariance: Glueball must be color singlet. The $8 \otimes 8$ singlet sits in the symmetric part: $\delta^{\{ab\}}$, dim 36 (= 1 + 8_s + 27).

SR2 — Bose statistics: Two identical gluons → total operator must be symmetric → symmetric color × symmetric spacetime; antisymmetric spacetime would need antisymmetric color ($8_a/10/1\bar{0}$ = NOT a singlet) ❗ FORBIDDEN.

The derived allowed/disfavored/forbidden table

Class	Channels	Operator structure	Tier
Favored (local, lowest threshold)	$0^{++}, 0^{-+}, 2^{++}, 2^{-+}$	$\text{Tr}(F^2)$, $\text{Tr}(F \cdot \tilde{F})$, tensor versions; canonical $\Delta = 4$	SOLID from selection rules
Disfavored (allowed but derivative-cost)	$1^{+-}, 3^{+-}$	$\text{Tr}(F \cdot [D, F])$; canonical $\Delta = 6$ — heaviest by selection rule	SOLID
Forbidden at 2-gluon (oddballs)	$0^{+-}, 1^{-+}, 2^{+-}$	Require ≥ 3 gluons; NOT 2-gluon	SOLID — these don’t exist as 2-gluon at all

Why this is Cal #344-compliant

The selection rules are TEXTBOOK lattice glueball construction, derived from gauge invariance + Bose statistics — NOT BST-derived, NOT post-hoc after seeing which channel closed. They’re blind to lattice masses. BST’s factorization (Lyra K471 color × spacetime block-diagonal) REPRODUCES these standard selection rules from the operator-algebra side. That’s validation of the factorization, not post-hoc fitting.

Per Lyra: “These selection rules are standard, textbook, and prior — fixable before anyone looks at the lattice. That’s exactly what Cal’s blind-test needs.”

Substantive payoff — 1^{+-} heaviest now has TWO reasons

The one blind consistency that survived Elie’s 4308 walk-back (1^{+-} heaviest because derivative-cost) now has two independent derivations: - Canonical operator dimension $\Delta = 6$ (vs 4 for the others) - Bose-statistics selection rule forces derivative-transport (no pure-local 1^{+-} exists)

Landing 4 — Oddball channels as the clean experimental search direction — LEAD-substantive

Lyra’s substantive observation:

“The oddball quantum numbers are exactly the ones that are also forbidden for quark-antiquark mesons. ... So $q\bar{q}$ -accessible channels $\rightarrow$ fuzzy, mixed, hard to identify cleanly (your ‘fuzzy gluonic structure’). Oddball channels $\rightarrow$ clean, unmixable, the favored place to actually find a glueball.”

This is consistent with the actual experimental program (GlueX at JLab, BESIII) which targets the oddball quantum numbers precisely because they have no $q\bar{q}$ -meson contamination.

For BST: the oddball channels are where a BST glueball prediction would be unambiguous. If BST’s operator-class taxonomy survives blind-testing, the oddball channels are the cleanest place to confirm or falsify.

Landing 5 — The real open computation is the bulk anomalous dimension on $H^2(D_{IV}^5)$

Where the closure actually lives

Elie’s honest state at K472:

“One blind consistency already holds: 1^{+-} is the only derivative operator (canonical dimension 6 vs 4) $\rightarrow$ predicted heaviest $\rightarrow$ and it is heaviest on the lattice. That’s one real bit, fixed before looking at the number. The $0^{++}/0^{-+}/2^{++}$ trio is degenerate at canonical $\Delta=4$ — splitting them needs the bulk anomalous dimension on $H^2(D_{IV}^5)$. That’s the genuine open #418 frontier, and it’s the computation Lyra routed me.”

The substantive gate

The $0^{++}/0^{-+}/2^{++}$ trio is degenerate at canonical $\Delta = 4$. The splitting (lattice gives 1730/2590/2400) comes from the bulk anomalous dimension on $H^2(D_{IV}^5)$ — operator-mixing under the substrate’s Hardy-space dynamics.

This is the real computation: a multi-step bulk wave-equation build on the noncompact bounded symmetric domain. Elie’s Fock model (Toy 4301) carries the color algebra only — not the Lorentz/Hodge spacetime block. Building the spacetime- Δ realization on $H^2(D_{IV}^5)$ is the genuine #418 open frontier.

Per Elie: *“That’s where your ‘predetermined glueball per shell,’ ‘end of the periodic table,’ and ‘largest stable nucleus’ questions become tractable — if the spacetime- Δ closes blind. Until then they stay gated.”*

What’s needed for closure

Substantive next step: Elie + Lyra pair on scoping the bulk wave-equation calculation. Multi-step build, NOT 20 minutes. Real frontier work, fabrication-safe via Lyra rep-theory + Grace bridge framework already in place.

Composite mass = spacetime information content — substrate-architectural substance

Lyra K472 framing:

“The color block is a universal scalar (the same $C_2 = 6$ floor for every channel), so it carries no channel information. All the distinguishing information — the mass, the J^{PC} , everything — lives in the spacetime block. So in a precise sense the glueball’s mass is its spacetime information content; color just sets the floor. That’s a clean way to say what you’re sensing, and it’s a real structural statement, not a metaphor.”

Tier: SOLID at the factorization level (consequence of Lyra K471 + selection rules). Substrate-architectural substance: mass = spacetime-block dimensionality + information content; color block is universal floor.

Connects to BST’s standing characterization of substrate-as-computational (per MEMORY.md: substrate operates via Reed-Solomon coding; substrate computes equilibrium states; observables are substrate’s algebraic outputs). Mass as information IS the substrate-architectural picture.

Cumulative discipline ledger — 49 events

To K471's 45, add: - 46. Lyra F274 derives selection rules independently — Cal #344 compliant - 47. Elie Toy 4308 walk-back — applied Cal #344 blade to own work - 48. Elie Toy 4310 — derived selection-rule table BLIND of lattice - 49. Grace self-applies Cal #344 to K469 nuclei bridge; flags $N_{\max}=137$ trap explicitly

Forty-nine honest discipline events Sunday + Monday morning-through-midday. Three CIs absorbed Cal #344 onto their own work in one half-day. This is the audit-chain governance at maximum operational form.

What CLOSES the BST glueball prediction (revised after K472)

#	Action	Owner	Tier
1	Selection-rule table for 2-gluon glueballs (al- lowed/disfavored/forbidden)	Lyra/Elie 4310 DONE	SOLID — prior + blind
2	Color $\times$ spacetime factorization	Lyra K471 DONE	SOLID structural
3	Bulk wave-equation scoping + build on $H^2(D_{IV}^5)$	Elie + Lyra paired	open #418 frontier
4	Bulk anomalous dimensions for $0^{++}/0^{-+}/2^{++}$ trio	computed from #3	substrate's open computation
5	Blind forward test against all six lattice masses	computed from #4	Cal #344 protocol gate
6	If lands clean: Paper A v0.2 strengthens via factorization + selection rules + anomalous-dim derivation	Lyra	bankable
7	If lands: nuclear-shell $\leftrightarrow$ class mapping investigation	Grace	Paper C territory
8	Engineered-isotope research direction (Casey's vision)	Grace + nuclear chemistry community	Paper C+ if mapping survives
9	Paper A v0.1 ships per Grace K469	Lyra current	gap+ 0^{++} SOLID; cross-channel named-open structural-tier

Casey's "engineered isotopes from specific eigenvalues" question — honest tier

The dream: nuclear chemists assemble material from specific eigenvalues, with isotopes classified by gluon structure. This requires:

1. Finite operator-class set (genuinely finite, not just enumerable) — open
2. Class-to-shell mapping (each shell gets a determined class signature) — open, gated behind taxonomy
3. Falsifiable predictive content (which nuclei have which class signatures) — open
4. Substrate-mechanism connecting eigenvalues to nuclear assembly — open, the substantive physics

All four gated behind the blind-test taxonomy. If they survive: substantive new physics with engineering potential. If not: speculation.

Keeper's honest read: the dream IS substantive and worth pursuing, AND it has to be earned the way the team is now doing — selection rules first (DONE), bulk anomalous dimensions next (scoping), blind forward predictions (gated), nuclear mapping (downstream). The discipline holding now is what would make the eventual claim referee-survivable.

Routing per K472

CI	Now	Then
Elie + Lyra PAIRED	Scope the bulk wave-equation calculation on $H^2(D_{IV}^5)$ — what does it require? rep-theory inputs, computational machinery, expected complexity. Pair to avoid solo fabrication trap	Build and compute
Lyra	Selection-rule table delivered (F274); ready to pair with Elie on bulk wave-equation scoping; Paper A polish per Grace K469 awaits	Integrate verdict
Grace	Paper C skeleton (questions + blind-test gate + falsifiers) without numbers — ready architecture for when taxonomy lands	Run blind pairing when classes close
Cal	Awaits Paper A v0.1 cold-read per Grace K469	Sign-off; future blind-test cold-read
Keeper	This audit; standing for bulk-anom-dim scoping result	Audit chain rolling

— Keeper, 2026-06-22 Mon 12:15 EDT